



Descriptive and Frequentist Statistics

Gayathri Jayan

M.Tech CSE – AI & Data Science
Cochin University of Science and Technology

2026

Full Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

What is Statistics?

Statistics is the branch of mathematics concerned with the collection, organization, analysis, interpretation, and presentation of data.

In simple words

Statistics helps us convert raw data into meaningful information for analysis and decision-making.

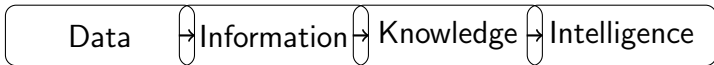
Two Major Areas

- **Descriptive Statistics** – summarizes and describes observed data.
- **Inferential Statistics** – uses sample data to draw conclusions about a population.

Frequentist Statistics

A framework for statistical inference based on samples and repeated-sampling ideas.

From Data to Intelligence



- **Data:** Raw facts and observations.
- **Information:** Organized and processed data.
- **Knowledge:** Understanding obtained from information.
- **Intelligence:** Using knowledge for reasoning and decisions.

Statistics and Artificial Intelligence

Artificial Intelligence (AI) aims to develop systems capable of performing tasks that normally require human-like intelligence.

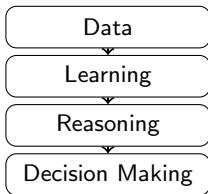
Examples of AI activities

- Learning from data
- Pattern recognition
- Reasoning
- Decision making
- Optimization
- Content generation

Role of Statistics

Statistics provides methods for understanding data, estimating parameters, measuring uncertainty, and evaluating models used in AI and Machine Learning.

Main Processes of AI



Important AI Processes

- **Decision Making** – selecting an appropriate action.
- **Optimization** – finding an effective solution.
- **Content Generation** – producing new content from learned patterns.

Outline

- Descriptive Statistics
 - Measures of Central Tendency
 - Measures of Dispersion
 - Frequentist Statistics
 - Model Estimation
 - Algebraic Structures and Vector Spaces
- Conclusion

Descriptive Statistics

Descriptive statistics consists of methods used to organize, summarize, and present data.

Numerical summaries

- Sample mean
- Sample variance
- Covariance
- Quantiles
- Order statistics

Graphical summaries

- Histograms
- Box plots
- Scatter plots
- Frequency distributions

Purpose

Descriptive statistics provides a compact representation of the important characteristics of observed data.

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

Measures of Central Tendency

Measures of central tendency describe the central or typical value of a dataset.

The three commonly used measures are:

- **Mean** – arithmetic average
- **Median** – middle value
- **Mode** – most frequently occurring value

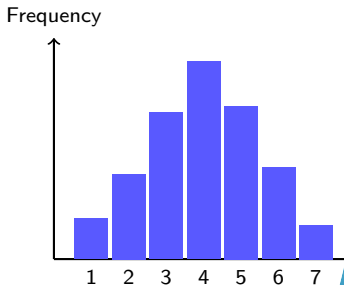
Purpose

They provide a single value that represents the center of the data.

Histogram

A **histogram** is a graphical representation of the distribution of numerical data.

- The data range is divided into intervals called **bins**.
- Each bar represents the frequency of observations within a bin.
- Histograms help identify the shape and spread of data.
- They can reveal skewness, concentration, and possible outliers.



Skewness

Skewness measures the asymmetry of a probability distribution around its mean.

$$\text{Skewness} = \frac{E[(X - \mu)^3]}{\sigma^3}$$

Positive Skew

- Longer tail on the right.
- Usually:

Mean > Median
Symmetric Distribution

For a symmetric distribution, skewness is approximately zero.

Negative Skew

- Longer tail on the left.
- Usually:

Mean < Median

Kurtosis

Kurtosis describes the tail behavior and peakedness of a distribution.

$$\text{Kurtosis} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

For normal distribution: Kurtosis = 3

Excess Kurtosis

$$\text{Excess Kurtosis} = \text{Kurtosis} - 3$$

- **Positive** excess kurtosis → heavier tails.
- **Negative** excess kurtosis → lighter tails.
- **Zero** excess kurtosis → same kurtosis as normal.

Types of Kurtosis

Type	Excess Kurtosis	Characteristic
Mesokurtic	$= 0$	Similar to normal distribution
Leptokurtic	> 0	Heavier tails and greater concentration around the center
Platykurtic	< 0	Lighter tails and flatter distribution

Key Idea

Kurtosis mainly describes the tail behavior of a distribution.

Sample Mean

Let $X_1, X_2, \dots, X_n$ be observations from a sample.
The **sample mean** is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Interpretation

Measures the central tendency of the observed data.

Example

For 2, 4, 6, 8, 10,

$$\bar{X} = \frac{30}{5} = 6$$

Mean, Median and Mode

Mean

Arithmetic average:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Sensitive to extreme values.

Median

Middle value after arranging the observations.

$$X_{(1)} \leq \dots \leq X_{(n)}$$

Less affected by outliers.

Mode

Most frequently occurring value.

Mode = most frequent value

May have more than one mode.

Example

For 2, 3, 3, 5, 7:

Mean = 4, Median = 3, Mode = 3

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

Measures of Dispersion

Measures of dispersion describe how spread out the observations are around a central value.

Important measures include:

- **Range**
- **Variance**
- **Standard Deviation**
- **Quartiles and Interquartile Range**

Interpretation

A small dispersion indicates that observations are close together, while a large dispersion indicates greater variability.

Sample Variance

Sample variance measures the dispersion of observations around the sample mean.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- Small variance $\rightarrow$ observations are close to the mean.
- Large variance $\rightarrow$ greater dispersion.
- $n - 1$ provides the degrees-of-freedom correction.

Standard Deviation

$$S = \sqrt{S^2}$$

It is expressed in the same units as the original data.

Standard Deviation

The **standard deviation** is the square root of the variance.

$$S = \sqrt{S^2}$$

For a sample:

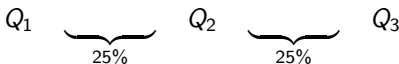
$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}$$

- Measures the typical spread around the mean.
- Smaller S indicates less variability.
- Larger S indicates greater variability.
- It has the **same units** as the original data.

Quartiles

Quartiles divide ordered data into four approximately equal parts.

- Q_1 – First quartile (25th percentile)
- Q_2 – Second quartile (50th percentile / median)
- Q_3 – Third quartile (75th percentile)



Interquartile Range

$$IQR = Q_3 - Q_1$$

Measures the spread of the middle 50% of observations.

Outliers

An **outlier** is an observation that lies unusually far from the majority of the data.

Using the IQR method:

$$IQR = Q_3 - Q_1$$

$$\text{Lower Fence} = Q_1 - 1.5(IQR)$$

$$\text{Upper Fence} = Q_3 + 1.5(IQR)$$

Outlier Rule

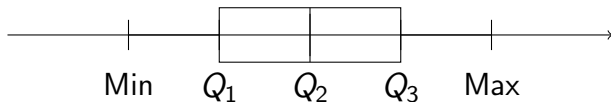
An observation is an outlier if

$$X < Q_1 - 1.5(IQR) \quad \text{or} \quad X > Q_3 + 1.5(IQR)$$

Box Plot and Outliers

A **box plot** summarizes the distribution using five values:

- Minimum
- Q_1
- Median (Q_2)
- Q_3
- Maximum



Key Idea

Points outside the whiskers can indicate potential outliers.

Order Statistics

Given a sample $X_1, X_2, \dots, X_n$, arrange the observations in increasing order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

- $X_{(1)}$ is the minimum.
- $X_{(n)}$ is the maximum.
- Middle order statistics determine the median.
- Quantiles are based on ordered observations.

Example

Original sample:

8, 2, 7, 4, 5

Ordered sample:

2, 4, 5, 7, 8

Median:

$$X_{(3)} = 5$$

Sample Covariance

Sample covariance measures how two variables vary together.

For variables X and Y ,

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Positive

Large values of X tend to occur with large values of Y .

Negative

Large values of X tend to occur with small values of Y .

Zero

Indicates little linear co-variation.

Sample Covariance Matrix

For p variables, the sample covariance matrix is

$$\mathbf{S} = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1p} \\ S_{21} & S_{22} & \cdots & S_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ S_{p1} & S_{p2} & \cdots & S_{pp} \end{bmatrix}$$

- Diagonal elements are sample variances.
- Off-diagonal elements are sample covariances.
- The covariance matrix is symmetric:

$$S_{ij} = S_{ji}.$$

Example

$$\mathbf{S} = \begin{bmatrix} 4 & 2 & -1 \\ 2 & 9 & 3 \\ -1 & 3 & 16 \end{bmatrix}$$

$$4, 9, 16$$

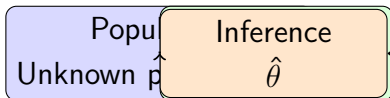
are the diagonal variances.

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- **Frequentist Statistics**
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

Frequentist Statistics

Frequentist statistics uses information from random samples to make conclusions about unknown population parameters.



Key Idea

Use sample data to estimate and make inferences about population parameters.

Sampling

Sampling is the process of selecting observations from a population.

Population

The complete collection of individuals, objects, or measurements of interest.

Parameter

A numerical characteristic of the population, such as μ or σ^2 .

Sample

A subset of observations selected from the population.

Statistic

A quantity calculated from the sample, such as $\bar{X}$ or S^2 .

Sampling Variability

Different random samples from the same population can produce different values of a statistic.

Types of Sampling

1. Simple Random Sampling

Every population member has an equal chance of selection.

2. Stratified Sampling

Population is divided into groups, and samples are taken from each group.

3. Systematic Sampling

Members are selected at regular intervals from an ordered population.

4. Cluster Sampling

Population is divided into clusters, and selected clusters are sampled.

Purpose

Study population characteristics using a manageable subset of observations.

Parameter and Statistic

Population Parameter

A numerical characteristic of a population.

μ = Population Mean

σ^2 = Population Variance

Sample Statistic

A numerical quantity calculated from sample data.

$\bar{X}$ = Sample Mean

S^2 = Sample Variance

Key Idea

Sample Statistic $\rightarrow$ Estimate of Population Parameter

Statistical Estimation

Estimation uses sample data to estimate an unknown population parameter.

Point Estimation

Provides a single value as an estimate.

$$\hat{\mu} = \bar{X} \quad \hat{\theta} = \text{estimate of } \theta$$

Interval Estimation

Provides a range of plausible values.

$$L < \theta < U$$

where L and U are the lower and upper limits.

Goal

Obtain accurate and reliable information about unknown population parameters.

Mean Square Error

Mean Square Error (MSE) measures the average squared difference between an estimator and the true parameter.

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Bias–Variance Decomposition:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + Bias(\hat{\theta})^2$$

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Interpretation

A smaller MSE indicates that an estimator tends to be closer to the true parameter.

Consistency

An estimator is **consistent** if it approaches the true population parameter as the sample size increases.

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

Intuition

Increasing sample size provides more information about the population.

Example

Under standard conditions, $\bar{X}$ is a consistent estimator of the population mean μ .

Note

Consistency is a large-sample property and is different from unbiasedness.

Confidence Intervals

A **confidence interval** provides a range of values from sample data for estimating an unknown population parameter.

For a population mean with known standard deviation:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- $\bar{X}$ – sample mean
- σ – population standard deviation
- n – sample size
- $z_{\alpha/2}$ – critical value

95% Confidence Level

Using the same procedure repeatedly, approximately 95% of the resulting intervals contain the true population parameter.

Interpretation of a 95% Confidence Interval

Suppose a 95% confidence interval is constructed for a population parameter θ .

$$L(X) < \theta < U(X)$$

Frequentist Interpretation

If samples are repeatedly taken and the same confidence procedure is applied, approximately 95% of the resulting intervals will contain the true parameter.

Important

The 95% refers to the **long-run performance of the procedure**, not a 95% probability assigned to the fixed parameter after the interval is calculated.

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- **Model Estimation**
- Algebraic Structures and Vector Spaces
- Conclusion

Parametric Model Estimation

Parametric estimation assumes that the population distribution belongs to a specified family described by a finite number of parameters.

Gaussian Model

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

Parameters:

$$\theta = (\mu, \sigma^2)$$

Common Methods

- Maximum Likelihood Estimation
- Method of Moments
- Bayesian Estimation

Advantage

When the assumed model is appropriate, parametric estimation can be efficient and require relatively little data.

Maximum Likelihood Estimation

Maximum Likelihood Estimation (MLE) chooses the parameter value that makes the observed data most likely.

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta)$$

where

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta)$$

- X_i – observed data
- θ – unknown parameters
- $L(\theta)$ – likelihood of the observed data

Goal

Find the parameter value that maximizes the likelihood of the observed data.

Maximum Likelihood and Log-Likelihood

The MLE is

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Since $\log(\cdot)$ is monotonically increasing,

$$\hat{\theta} = \arg \max_{\theta} \log L(\theta)$$

Why use Log-Likelihood?

- Converts products into sums.
- Simplifies differentiation.
- Improves numerical stability.

$$\log L(\theta) = \log \left(\prod_{i=1}^n f(X_i; \theta) \right) = \sum_{i=1}^n \log f(X_i; \theta)$$

Non-Parametric Model Estimation

Non-parametric estimation makes fewer assumptions about the underlying probability distribution.

Characteristics

- Flexible model structure
- Fewer distributional assumptions
- Models complex distributions
- Often requires more observations

Examples

- Histogram
- Empirical distribution function
- Kernel density estimation
- Rank-based methods

Trade-off

Greater flexibility can require more data and lead to more complex estimation.

Kernel Density Estimation

Kernel Density Estimation (KDE) is a non-parametric method for estimating the probability density function from observed data.

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

- n – number of observations
- h – bandwidth
- K – kernel function
- X_i – observed data points

Bandwidth

Small h gives a more detailed estimate; large h gives a smoother estimate.

Parametric vs. Non-Parametric Estimation

Aspect	Parametric	Non-Parametric
Distribution	Specified family	Flexible / unspecified
Assumptions	Stronger	Fewer
Parameters	Finite number	Flexible complexity
Flexibility	Lower	Higher
Data requirement	Often lower	Often higher
Examples	Gaussian model, MLE	Histogram, KDE

Selection

Parametric methods are useful when model assumptions are reasonable; non-parametric methods are useful when the distributional form is uncertain.

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

Magma

A **magma** is a non-empty set equipped with a binary operation that is closed on the set.

Let M be a non-empty set and

$$* : M \times M \rightarrow M.$$

Then $(M, *)$ is a magma if

$$a * b \in M \quad \forall a, b \in M$$

Key Property

Closure under the binary operation.

Example

$(\mathbb{Z}, +)$ is a magma because

$$a + b \in \mathbb{Z} \quad \forall a, b \in \mathbb{Z}.$$

Semigroup

A **semigroup** is a magma whose binary operation is **associative**.

For a set S with operation $*$:

$$(a * b) * c = a * (b * c)$$

for all $a, b, c \in S$.

A semigroup therefore satisfies:

1. Closure
2. Associativity

Example

$(\mathbb{N}, +)$ is a semigroup because addition is closed and associative.

Monoid

A **monoid** is a semigroup that contains an **identity element**.
For $(M, *)$, there exists $e \in M$ such that

$$e * a = a * e = a \quad \forall a \in M$$

A monoid satisfies:

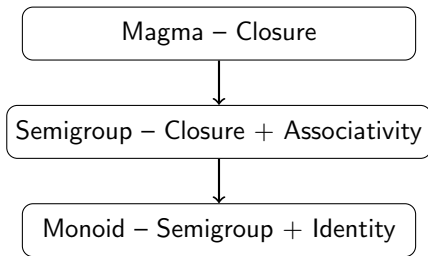
1. Closure
2. Associativity
3. Identity element

Example

$(\mathbb{N}_0, +)$ is a monoid with identity

$$e = 0.$$

Magma, Semigroup and Monoid



Hierarchy

Every monoid is a semigroup, and every semigroup is a magma.

Vector Space

A **vector space** is a set equipped with vector addition and scalar multiplication.

Let V be a vector space over a field $\mathbb{F}$:

$$+ : V \times V \rightarrow V, \quad \cdot : \mathbb{F} \times V \rightarrow V.$$

For $u, v \in V$ and $a, b \in \mathbb{F}$:

- $u + v \in V, \quad av \in V$
- Addition is associative and commutative
- Zero vector and additive inverses exist
- Scalar multiplication satisfies distributive and associative laws

Example of a Vector Space

Consider

$$V = \mathbb{R}^2, \quad v = \begin{bmatrix} x \\ y \end{bmatrix}.$$

For

$$u = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix},$$

Vector Addition:

$$u + v = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix}$$

Scalar Multiplication:

$$av = \begin{bmatrix} ax \\ ay \end{bmatrix}$$

Examples

$\mathbb{R}^2$ and $\mathbb{R}^3$ are standard vector spaces.

Hyperplane

A **hyperplane** is a flat affine subspace of dimension one less than its surrounding space.

In $\mathbb{R}^n$:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the coefficients are not all zero.

In $\mathbb{R}^2$:

$$ax + by = c$$

represents a **line**.

In $\mathbb{R}^3$:

$$ax + by + cz = d$$

represents a **plane**.

Key Idea

A hyperplane divides the space into two half-spaces.

Hyperplane in Machine Learning

In machine learning, a hyperplane is commonly used as a **decision boundary**.

$$w^T x + b = 0$$

where w is the weight vector, x is the input vector, and b is the bias.

$$w^T x + b > 0$$

⇒ **Class 1**

$$w^T x + b < 0$$

⇒ **Class 2**

Application

Hyperplanes form the basis of linear classifiers such as Support Vector Machines (SVM).

Outline

- Descriptive Statistics
- Measures of Central Tendency
- Measures of Dispersion
- Frequentist Statistics
- Model Estimation
- Algebraic Structures and Vector Spaces
- Conclusion

Conclusion

- **Descriptive statistics** summarizes data using measures of center, spread, and distribution shape.
- **Histograms, box plots, IQR, skewness, and kurtosis** help understand distributions and identify outliers.
- **Covariance** describes relationships between variables.
- **Frequentist statistics** uses samples to make inferences about populations.
- **Estimation, MSE, consistency, and confidence intervals** are important tools for statistical inference.
- **Parametric and non-parametric methods** provide different approaches to statistical modelling.
- These statistical concepts form an important foundation for **AI and Machine Learning**.

Key Takeaway

Statistics transforms data into reliable information for analysis, inference, and intelligent decision-making.



Thank You